INVESTIGATING THE COLLECTIVE MECHANISM OF POLYBIA-MP1 ACROSS MEMBRANE MODELS

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Keywords: Model Membranes, Peptides, Molecular Dynamics

ABSTRACT

Antimicrobial peptides are short, predominantly cationic molecules whose mechanism of action involves electrostatic and hydrophobic interactions with cell membranes, disrupting lipid packing and inducing pores or defects that lead to cell death. Polybia-MP1 (MP1) is a peptide that is well known for its selective action against bacteria, targeting anionic phospholipids like phosphatidylglycerol (PG). However, the molecular details of its collective mechanism, specifically regarding pore formation and the impact of membrane complexity, remain to be fully elucidated. We employ a multiscale computational strategy to investigate the collective behavior of MP1 in different membrane models. Preliminary All-Atom (AA) molecular dynamics simulations characterize the initial adsorption of individual peptides, while Coarse-Grained (CG) simulations overcome the temporal and spatial limitations to capture concentration-dependent events, including peptide aggregation and the impact of the peptides on different membrane compositions. We aim to identify the structural features that optimize lytic action and selectivity, bridging the gap between specific atomic interactions and large-scale membrane processes.

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